



13

13 



Easy Cryptography picoCTF 2019

AUTHOR: ALEX FULTON/DANIEL TUNITIS

Description

Cryptography can be easy, do you know what ROT13 is?
cvpbPGS{abg_gbb_onq_bs_n_ceboy rz}

Hints 

1

This can be solved online if you don't want to do it by hand!

103,195 users solved



88%



Liked

 picoCTF{FLAG}

Submit
Flag

Author: The Analyst: Hyposelenia

Challenge: 13

Category: Cryptography

Date: 02/24/26



I. Objective

The objective of the “13” challenge was to decode an encoded message and retrieve the hidden flag using basic cryptographic knowledge.

II. Background

What is Cryptography?

Imagine you want to send a secret message to your friend. But you don't want other people to understand it.

So, you:

- Mix the letters
- Change the letters
- Hide the real words

That is **cryptography**.

Cryptography is the practice of:

- Hiding information
- Protecting messages
- Making secrets unreadable to others

Long ago, people used it to:

- Send secret war messages
- Protect important information
- Communicate privately

Today, it is used in:

- Passwords
- Online banking
- Secure websites
- Encrypted messaging



What is a Caesar Cipher?

A **Caesar Cipher** is one of the **oldest encryption methods**.

It works by:

- Shifting each letter forward in the alphabet by a certain number.

Example (Shift by 3):

A → D

B → E

C → F

If you shift the word:

DOG

By 3 letters, it becomes:

GRJ

It is named after **Julius Caesar**, who reportedly used it to send secret military messages.

What is ROT13?

ROT13 stands for:

Rotate by 13 letters.

It is a special type of Caesar Cipher where:

- Each letter is shifted **13 positions**
- The alphabet has 26 letters
- 13 is exactly half of 26

That means:

If you apply ROT13 twice, you get the original text back.



Example:

A ↔ **N**

B ↔ **O**

C ↔ **P**

So:

HELLO becomes **URYyb**

And if you decode **URYyb** using ROT13 again, you get **HELLO**.

III. Tool Used

- **ROT13 Online Decoder** - Used to decode rotated 13 texts

IV. Methodology

1. Analyzed the given text in the challenge description.

13 

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Cryptography can be easy, do you know what ROT13 is?

cvpbPGS{abg_gbb_onq_bs_n_ceboyrz}



2. Noticed it resembled a picoCTF flag but did not start with “picoCTF{”.

cvpbPGS{abg_gbb_onq_bs_n_ceboy rz}

3. Since the category was **Cryptography**, assumed it was encrypted.



4. Based on the title “13”, suspected ROT13.
5. Searched for an online ROT13 decoder.
6. Inputted the ciphertext.



7. Retrieved the plaintext, which was the correct flag.

V. Result

picoCTF{not_too_bad_of_a_problem}



VIEW

Plaintext ▼

```
picoCTF{not_too_bad_of_a_problem}
```

VI. Explanation

This challenge was a basic demonstration of:

- Classical encryption
- Pattern recognition
- Understanding cipher clues

The title “13” was the main hint.

Since:

- The text looked like a flag
- The category was cryptography
- The number 13 was emphasized

It strongly suggested **ROT13**.

ROT13 is not secure encryption. It is simply letter rotation.



This type of cipher was created mainly to:

- Hide messages casually
- Prevent casual reading
- Add light secrecy
-

It is very easy to break, but it teaches an important lesson:
Even simple encryption changes how information looks — and recognizing patterns is key in cryptography.

VII. Conclusion

The **13** challenge reinforces foundational cryptographic concepts:

- Understanding Caesar ciphers
- Recognizing encoding patterns
- Using contextual clues to identify encryption methods

By identifying the cipher type and decoding the message, the hidden flag was successfully retrieved.

This challenge serves as an introduction to classical cryptography and highlights the importance of analyzing patterns when solving encoded messages.

— **The Analyst: Hyposelenia**